Basics of Statistics

The building blocks: data, variables, scales and frequency distributions.

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1 What is statistics?

Statistics is the science which deals with the **collection, organisation (classification), presentation, analysis and interpretation of data**.

- **Origin:** Italian *statista* (statesman), German *Statistik*, used by **Gottfried Achenwall** (1749).
- **Sir John Sinclair** brought the word into English through the *Statistical Account of Scotland* (1791–99, 21 volumes).
- **Sir R.A. Fisher** (1890–1962): **Father of Modern Statistics**. Gave us ANOVA, experimental design and maximum likelihood, and the principles of **randomisation, replication and local control**.

2 Two branches

Statistics

Descriptive	Inferential
Summarises a sample or population with measures such as the mean or standard deviation.	Uses a random sample to describe and draw inferences about population parameters.

Data are individual pieces of factual information: the raw material of statistics (e.g. rainfall over a period, area under paddy in a state).

3 Statistics vs maths

	Mathematics	Statistics
Reasoning	Deductive	Inductive
Results	Exact, unchanging	Under uncertainty
Handles	Axioms, proofs	Variable real-world data

The **error term** captures the variability we cannot control: observation − true model mean.

4 Population, sample, parameter

Population
The entire collection of units with the characteristic under study.
e.g. heights of **all** students in a college

Sample
A subset selected for study; should be **representative** of the population.
e.g. heights of 20 selected students

Parameter
A numerical characteristic of a population. It is **fixed** but usually **unknown**.
$\mu,\ \sigma^2,\ P$

Statistic
A value calculated from sample observations; used to **estimate** the parameter.
$\bar{x},\ s^2,\ \hat{p}$

5 Variables & constants

Variable: takes different values for different individuals, times or places.
Constant: stays fixed (e.g. $\pi = 3.14159\ldots$, $e = 2.71828\ldots$).

Type	Meaning	Agri examples
Discrete quantitative	Countable, separate values (0, 1, 2 …)	Fruits per branch, plots in a field
Continuous quantitative	Any value in a range	Plant height, yield, panicle length
Categorical qualitative	Categories, not measurements; can be counted	Variety, colour, disease present/absent

6 Measurement scales

Scale	Order	Equal gaps	True zero	Example
Nominal	×	×	×	Blood group, gender
Ordinal	✓	×	×	Grades (excellent, good …)
Interval	✓	✓	×	Temperature in °C
Ratio	✓	✓	✓	Weight, Kelvin

40 °C is **not** "twice as hot" as 20 °C (no true zero), but 60 kg **is** twice 30 kg.

7 Collection of data

Primary data
First-hand, collected for the study's own purpose; no prior statistical treatment.
e.g. population census by the government

Secondary data
Already collected and processed by others for another purpose.
e.g. Economic Survey, Ministry reports

PRIMARY METHODS personal investigation · trained enumerators · correspondents · phone or mailed questionnaire · online forms · mobile apps and social media polls

SECONDARY SOURCES official (RBI, ministries, NSO) · institutional (universities, research institutes). **Enumerator** collects; **respondent** provides.

8 Frequency distribution terms

Class limits	End points of a class, e.g. 110–119
Class boundaries	Close the gaps: 109.5–119.5 for data recorded to the nearest cm
Continuous classes	No gaps (10–20, 20–30): limits = boundaries; use $10 \le x < 20$
Class mark	Mid-point of the class interval
Class width	Upper boundary − lower boundary
LCF / GCF	Less-than (at upper boundaries) / greater-than (at lower boundaries) running totals
Relative freq.	f/N; all of them add up to 1

9 Building a grouped table

1. Decide the number of classes k
2. Determine the class width
3. Set up the class limits
4. Tally items into classes (every 5th stroke crosses the first four)
5. Count the frequency in each class

Raw data → **ungrouped** table (each value) or **grouped** table (classes). Grouping gives a clear picture but loses individual values.

10 Worked example · leaf length (cm), N = 38

Sturges: $k = 1 + 3.322 \log_{10} 38 \approx 6.25 \Rightarrow$ 6 classes; width $(18-1)/6 \approx 2.83 \Rightarrow 3$ cm.

Class	Boundaries	Mark	f	LCF	GCF	RF
1–3	0.5–3.5	2	3	3	38	0.079
4–6	3.5–6.5	5	6	9	35	0.158
7–9	6.5–9.5	8	10	19	29	0.263
10–12	9.5–12.5	11	5	24	19	0.132
13–15	12.5–15.5	14	5	29	14	0.132
16–18	15.5–18.5	17	9	38	9	0.237

11 Formulae to remember

Number of classes (Sturges)

$$k = 1 + 3.322 \log_{10} N$$

Approximate class width

$$C = \frac{|\max - \min|}{k}$$

Lower boundary of class i

$$L_i = \frac{\mathrm{LCL}_i + \mathrm{UCL}_{i-1}}{2}$$

Class mark

$$\frac{\text{Lower limit} + \text{Upper limit}}{2}$$

Relative frequency

$$\mathrm{RF} = \frac{f}{N}, \quad \sum \frac{f}{N} = 1$$

Sum of frequencies

$$\sum f = N$$

LCL / UCL = lower / upper class limit. Sturges' k is a guide, not a compulsory number.

! Common mistakes

- × A large dataset cannot correct biased selection or poor measurement.
- × Association alone does not prove cause and effect.
- × Online forms and social media polls collect voluntary responses, so they are not automatically representative.
- × Classes must cover the data **without overlap**. Boundaries depend on the recording precision.

? Fill in the blanks

1. Weight of a harvested mango recorded as 152.4 g is a _____ variable. **[continuous]**
2. The average yield of 100 sampled paddy farms is a _____; the true average of every farm in the district is a _____. **[statistic; parameter]**
3. Temperature in °C is on an _____ scale, so a 40 °C day is not "twice as hot" as a 20 °C day. **[interval]**
4. Rating farmers' satisfaction as low, medium or high uses the _____ scale. **[ordinal]**
5. A researcher using rainfall figures copied from an IMD report is working with _____ data. **[secondary]**
6. Plant height recorded to the nearest cm in the class 110–119 has class boundaries _____ and _____. **[109.5 and 119.5]**

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Official Agricultural Statistics

How India measures crop area, yield and production, and what the Agricultural Census records.

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1 Who collects what

- Agriculture is a **state subject**, but agricultural statistics are on the **concurrent list**, so the system is **decentralised**.
- **State Agricultural Statistics Authorities (SASAs)** collect and compile at state level.
- Nationally, the **ES&E Division** (Economics, Statistics and Evaluation) of the Dept. of Agriculture & Farmers Welfare compiles crop production estimates.

Crop production

$$\text{Production} = \text{Area} \times \text{Average yield}$$

2 Area statistics: three systems

System	Where	Share*
Complete enumeration	Temporarily settled states with cadastral land records	~86%
Sample surveys	Permanently settled states with no village revenue agency, e.g. **Kerala**, Odisha, West Bengal	~9%
No developed system	Hill and north-eastern states, some UTs (personal assessment)	~5%

*Share of the country's reporting area, ICS report 2018–19.

3 Crop Cutting Experiment (CCE): steps

A small, randomly located plot is harvested and weighed to estimate yield. Yield surveys for principal crops run under the **General Crop Estimation Survey (GCES)**.

1. **Select the field:** it must be larger than the CCE plot.
2. **Locate the plot** from the **south-west corner** using random steps along the horizontal and vertical sides.
3. **Mark and harvest** plants rooted inside the pegged, string-marked plot; apply the boundary rule consistently.
4. **Thresh** on a clean mat and dry the grain.
5. **Winnow and weigh** : subtract the empty bag to get the net grain weight.

4 CCE numbers

Check a square plot

$$d = a\sqrt{2};\ 5\ \text{m plot} \Rightarrow 7.07\ \text{m}$$

Plot yield (kg/ha)

$$\frac{10\,000\,w}{a}$$

w = net grain (kg), a = plot area (m²). 12.5 kg from 25 m² = 5000 kg/ha.

Plot shapes: 5×5 m, 10×10 m squares, 10×5 m rectangle; triangles (UP) and circles (WB) also documented.

5 GCES sampling design

Strata

Stage 1	Stage 2	Stage 3
Villages	Fields	CCE plots

A **multistage** design. Plots are chosen by the prescribed random procedure, never by convenience. One plot does not represent a whole region.

6 Production estimates released

Release	Usually	Covers
First Advance	September	Kharif
Second Advance	February	Kharif + Rabi
Third Advance	May	Updated area and yield
Final	Sept / Oct	Final estimates

Advance estimates exist because final figures come only after harvest. Quote any estimate with its year and release status.

7 Agricultural Census

What

Structure of agriculture: number and area of **operational holdings**, land use, cropping pattern, inputs, down to village level.

When

Started **1970–71**; **quinquennial** (every 5 years). 11th census: reference year **2021–22**, launched 28 July 2022.

How

Land-record states: number and area by **complete enumeration**; characteristics on a **sample basis**.

Reference period

The agricultural year, **July to June**.

! Common mistakes

- × Kerala uses **sample surveys** for area, not complete enumeration.
- × Yield = production ÷ **area**, never ÷ number of farmers.
- × Don't mix up the current three advance estimates with older four-forecast schedules.

? Fill in the blanks

1. A district with 20 000 ha under paddy and an average yield of 4 t/ha produces ——— t. [80 000]
2. 12.5 kg of grain from a 25 m² CCE plot is a yield of ——— kg/ha. [5000]
3. Each diagonal of a 10 m × 10 m CCE plot should measure about ——— m. [14.14]
4. Kerala has no village revenue agency, so its crop area is estimated by ———. [sample surveys]
5. Kharif production first appears in the ——— Advance Estimates. [First]
6. The Agricultural Census is quinquennial, so the census after reference year 2021–22 refers to ———. [2026–27]

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Graphical Representation

Histograms, ogives, polygons, stem-and-leaf plots, bar and pie charts, and boxplots.

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1 Histogram

Adjoining rectangles for a **continuous** frequency distribution; class boundaries on the x-axis.

Equal widths	Unequal widths
Bar height = **frequency**	Bar height = **frequency density**

Frequency density

$$\text{FD} = \frac{\text{Class frequency}}{\text{Class width}}$$

The mode can be estimated graphically from the histogram.

2 Ogives (cumulative frequency curves)

Type	Plot	Shape
Less-than	LCF against **upper** class boundaries	Rises
Greater-than	GCF against **lower** class boundaries	Falls

The two ogives **intersect at the median**. The median can also be read from the less-than ogive at $N/2$.

3 Frequency polygon

Plot frequencies at **class midpoints** and join them with straight lines. Useful for comparing two distributions on one graph.

4 Stem-and-leaf plot

Each value split into a **stem** (leading digits) and a **leaf** (trailing digit). Shows the shape **and keeps every observation**.

e.g. 107 → stem 10, leaf 7.

5 Bar charts

- **Simple**: one value per category
- **Component** (sub-divided): parts of each category
- **Grouped** (multiple): several values within each category

For categorical data, so bars are separated by gaps.

6 Histogram vs bar chart

	Histogram	Bar chart
Data	Continuous quantitative	Categorical / discrete
Bars	Adjacent, no gaps	Separated by gaps
Area	Represents frequency	Only height matters
Order	Fixed by the scale	Categories can be reordered

7 Pie chart

A circle split into sectors showing **parts of a whole**. The full circle is **360°**.

Sector angle

$$\frac{\text{Value of category}}{\text{Sum of all values}} \times 360^\circ$$

Best with a few categories that add up to a meaningful total.

8 Boxplot (box-and-whisker)

Shows the **five-number summary**: minimum, Q_1, **median** (line inside the box), Q_3, maximum, plus the spread and potential outliers.

Interquartile range

$$IQR = Q_3 - Q_1$$

Lower fence

$$Q_1 - 1.5 \times IQR$$

Upper fence

$$Q_3 + 1.5 \times IQR$$

Points beyond the fences are flagged as **potential outliers**.

! Common mistakes

- × Unequal class widths with raw frequencies as heights.
- × Plotting the less-than ogive at **lower** boundaries.
- × Leaving gaps between histogram bars.
- × Pie charts for data that are not parts of a whole.

? Fill in the blanks

1. Classes 0–10 and 10–30 each have frequency 20; their histogram bar heights should be _____ and _____. **[2 and 1 (frequency densities)]**
2. A crop occupying 25% of the cropped area gets a pie-chart sector of _____°. **[90]**
3. To read the median from a less-than ogive with N = 60, draw a horizontal line from cumulative frequency _____. **[30]**
4. If Q_1 = 20 and Q_3 = 30, any observation above _____ is flagged as a potential outlier. **[45]**
5. Area under three crops in each district, stacked inside one bar per district, is shown by a _____ bar chart. **[component]**
6. A display that shows the shape of the data yet lets you recover every observation is the _____ plot. **[stem-and-leaf]**

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Central Tendency I

Simple averages: the arithmetic mean, the median and the mode.

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1 The idea · types of averages

A measure of central tendency is a **single value that represents the centre or typical value** of a dataset.

Simple averages this chapter	Mean · Median · Mode
Special averages	Geometric mean · Harmonic mean
Positional averages	Quartiles · Deciles · Percentiles

Objectives: condense data into a single value; make comparisons easy.

2 A good average should be...

- ✓ Rigidly defined
- ✓ Simple to understand and easy to calculate
- ✓ Based on **all** values of the data
- ✓ Capable of further mathematical treatment
- ✓ Stable from sample to sample (sampling stability)
- ✓ Not unduly affected by extreme values

No single average meets all six. The mean fails only the last one.

3 Arithmetic mean

The sum of the observations divided by their number. It is what people usually mean by "average".

Raw data

$$\bar{x} = \frac{\sum x_i}{n}$$

Frequency distribution

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

Assumed mean, $d_i = x_i - A$

$$\bar{x} = A + \frac{\sum f_i d_i}{\sum f_i}$$

Coding, $u_i = (x_i - A)/C$

$$\bar{x} = A + C\,\frac{\sum f_i u_i}{\sum f_i}$$

Weighted mean

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

Combined mean of groups

$$\bar{x} = \frac{\sum n_j \bar{x}_j}{\sum n_j}$$

For grouped data, x_i is the class mark, so the mean is an approximation. With equal weights, the weighted mean is the simple mean.

4 Properties of the mean

Deviations sum to zero

$$\sum (x_i - \bar{x}) = 0$$

Least squares

$$\sum (x_i - a)^2 \text{ is least at } a = \bar{x}$$

Change of origin & scale

$$y_i = a + bx_i \Rightarrow \bar{y} = a + b\bar{x}$$

Combine group means by weighting with group sizes, not by averaging the means.

5 Median

The **middle value** after arranging the data in ascending order. It is a **positional** average.

Odd n

$\left(\frac{n+1}{2}\right)^{\text{th}}$ observation

Even n

mean of the $\left(\frac{n}{2}\right)^{\text{th}}$ and $\left(\frac{n}{2}+1\right)^{\text{th}}$ observations

Grouped data (linear interpolation)

$$Md = L + \left(\frac{\frac{N}{2} - F}{f_m}\right) C$$

Median class holds the $(N/2)^{\text{th}}$ value · L its lower boundary · F cumulative frequency below L · f_m its frequency · C its width. Graphically, read it off the less-than ogive at $N/2$.

6 Mode

The value that **occurs most often**. Found by inspection for raw data. It is the right average for **qualitative** (nominal) data.

Grouped data (equal class widths)

$$Mo = L + \left(\frac{f_m - f_p}{2f_m - f_p - f_s}\right) C$$

Modal class has the highest frequency · L its lower boundary · f_p, f_s frequencies of the preceding and succeeding classes · C class width. Graphically, use the histogram.

With **unequal widths**, compare frequency densities rather than raw frequencies. Data can be **bimodal** or have no mode.

7 Mean vs median vs mode

	Mean	Median	Mode
Based on	All values (calculated)	Position	Frequency
Uses every observation	Yes	No	No
Extreme values	Affected very much	Not influenced	Not affected
Open-end classes	Cannot compute	If median class is known	Modal category, yes
From a graph	No	Ogive	Histogram
Qualitative data	No	Ordinal only	Yes
Always unique	Yes	Yes	May be several or none

8 Shape & order

- **Symmetric**, unimodal: mean = median = mode
- Many **positively skewed** distributions: mean > median > mode
- Many **negatively skewed** distributions: mean < median < mode

The skewed orderings are typical, **not** a universal theorem.

Positive observations (preview)

$$AM \geq GM \geq HM$$

9 Worked results from the chapter

Problem	Working	Answer
Mean of 2, 4, 7, 8, 11, 12	$44/6$	**7.33**
Weighted mean, x = 10, 20, 30; w = 1, 2, 3	$(10 + 40 + 90)/6$	**23.33**
Median height, N = 60, median class 160–165	$160 + \frac{30-28}{18} \times 5$	**160.56 cm**
Mode of sorghum ear-head weight, modal class 100–120	$100 + \frac{45-38}{90-38-35} \times 20$	**108.24 g**

! Common mistakes

- × Averaging group means without weighting by group size.
- × Using class **limits** instead of **boundaries** for L.
- × Forgetting to sort the data before finding the median.
- × Applying the mode formula to unequal class widths.

? Fill in the blanks

1. The mean yield of 5 plots is 12 q, so their total yield is ——— q. [60]
2. If 3 cm is added to every plant height, the mean height increases by ——— cm. [3]
3. The deviations of 4, 6 and 8 from their mean add up to ———. [0]
4. For 7 observations arranged in order, the median is the ——— observation. [4th]
5. For grain colour recorded as red, white or brown, the only meaningful average is the ———. [mode]
6. Group A (10 plots, mean 20 q) and group B (30 plots, mean 40 q) have a combined mean of ——— q. [35]

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Central Tendency II

Special averages (geometric and harmonic means) and positional averages (quartiles, deciles, percentiles).

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1 Geometric mean (GM)

The nth root of the product of n positive observations. Use it for **growth rates, ratios, percentages** and multiplicative change.

Definition

$$GM = \sqrt[n]{\prod x_i}$$

Using logs

$$GM = \text{Antilog}\left(\frac{\sum \log x_i}{n}\right)$$

Frequency distribution

$$GM = \text{Antilog}\left(\frac{\sum f_i \log x_i}{n}\right)$$

Needs positive values: a zero makes the product GM zero, and it is unsuitable for negatives.

2 Harmonic mean (HM)

The **reciprocal of the arithmetic mean of the reciprocals**. Use it for **rates**, such as average speed over equal distances.

Raw data

$$HM = \frac{n}{\sum \frac{1}{x_i}}$$

Frequency distribution

$$HM = \frac{n}{\sum \frac{f_i}{x_i}}$$

GM and HM both give **more weight to smaller values**.

3 AM · GM · HM

Together: the **Pythagorean means**.

Positive observations

$$AM \geq GM \geq HM$$

Equal only when all observations are equal.

Two positive values

$$AM \times HM = GM^2$$

4 Which mean?

Mean	Use for	Agri example
AM	Additive quantities, general averages	Average yield, rainfall
GM	Growth factors, ratios, percentages	Compound growth of area or production
HM	Rates and reciprocal relationships	Average speed of a tractor over equal distances

5 Positional averages

Measure	Divides into	Key equalities
Quartiles Q_1, Q_2, Q_3	4 parts	Q_2 = Median
Deciles $D_1 \dots D_9$	10 parts	D_5 = Median
Percentiles $P_1 \dots P_{99}$	100 parts	$P_{25} = Q_1,\ P_{50} = Q_2,\ P_{75} = Q_3$

Raw data: **sort first**. Discrete data: use cumulative frequencies. Grouped data: find the class, then interpolate.

6 Raw data positions

First quartile

$$Q_1 = \left(\tfrac{n+1}{4}\right)^{\text{th}}$$

Third quartile

$$Q_3 = \left(\tfrac{3(n+1)}{4}\right)^{\text{th}}$$

dth decile

$$D_d = \left(\tfrac{d(n+1)}{10}\right)^{\text{th}}$$

pth percentile

$$P_p = \left(\tfrac{p(n+1)}{100}\right)^{\text{th}}$$

7 Grouped data (interpolation)

Quartile Q_1

$$Q_1 = l_1 + \frac{\frac{n}{4} - m_1}{f_1}\, c_1$$

Quartile Q_3

$$Q_3 = l_3 + \frac{\frac{3n}{4} - m_3}{f_3}\, c_3$$

Decile

$$D_d = l + \frac{\frac{dn}{10} - cf}{f}\, c$$

Percentile

$$P_p = l + \frac{\frac{pn}{100} - cf}{f}\, c$$

l lower boundary of the class holding the position · m or cf cumulative frequency before it · f its frequency · c its width.

! Common mistakes

- × Forgetting the **antilog** at the end of a GM.
- × Using GM or HM with zero or negative values.
- × AM for average speed over equal distances (use HM).
- × Finding quartiles before sorting.

? Fill in the blanks

1. A tractor ploughs two equal distances at 4 km/h and 6 km/h; its average speed is ——— km/h. **[4.8 (harmonic mean)]**
2. Area grew by factors 2 and 8 in two successive years; the average growth factor is ———. **[4 (geometric mean)]**
3. Two positive numbers have AM = 10 and HM = 6.4, so their GM is ———. **[8]**
4. The 50th percentile, the 5th decile and the 2nd quartile are all equal to the ———. **[median]**
5. The 75th percentile is the same as the ——— quartile. **[third]**
6. The geometric mean cannot be used if any observation is ———. **[zero or negative]**

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Measures of Dispersion

Range, interquartile range, mean absolute deviation, variance, standard deviation and the coefficient of variation.

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1 What is dispersion?

The **extent to which observations deviate from a central value**: the spread of the data.

Absolute measures	Relative measures
In the data's units: range, IQR, MAD, SD	Unit-free: coefficient of variation (CV)

A good measure is rigidly defined, easy to compute, based on all values, suitable for further algebra and not unduly affected by extremes.

2 Range & IQR

Range

$$R = L - S$$

Interquartile range

$$IQR = Q_3 - Q_1$$

- **Range**: simplest; uses only the two extremes.
- **IQR**: spread of the **middle 50%**; less affected by extremes.
- **Quartile deviation** = IQR ÷ 2.

3 Mean absolute deviation

MAD about the mean

$$MAD = \frac{\sum |x_i - \bar{x}|}{n}$$

Uses all observations but ignores signs. MAD is **least about the median**.

4 Variance & standard deviation

Sample variance

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

Shortcut

$$s^2 = \frac{1}{n-1}\left\{\sum x_i^2 - \frac{(\sum x_i)^2}{n}\right\}$$

Discrete frequency

$$s^2 = \frac{1}{n-1}\left\{\sum f_i x_i^2 - \frac{(\sum f_i x_i)^2}{n}\right\}$$

Grouped, $d_i = (x_i - A)/c$

$$s = c\sqrt{\frac{1}{n-1}\left\{\sum f_i d_i^2 - \frac{(\sum f_i d_i)^2}{n}\right\}}$$

SD s = non-negative square root of s^2. Grouped data: x_i = class mark, A = class mark with the highest frequency, c = class interval.

5 Properties to remember

- ✓ Variance is in **squared** units; SD is in the **original** units.
- ✓ SD is always ≥ 0; SD = 0 means all observations are identical.
- ✓ Sample variance uses divisor $n - 1$ (unbiased).
- ✓ Adding a constant to every value does **not** change SD.
- ✓ Multiplying by b multiplies SD by $|b|$.
- ✓ Squaring gives extreme deviations more weight.

6 Coefficient of variation

CV (%)

$$CV = \frac{s}{\bar{x}} \times 100$$

A **relative** measure, used to compare variability of datasets in **different units or with different means**.

Higher CV	Lower CV
More relative variability: less consistent	Less relative variability: more consistent

7 Comparing the measures

Measure	Uses	Extreme values	Note
Range	2 extremes only	Strongly affected	Quick, rough
IQR	Middle 50%	Resistant	Pairs with the median
MAD	All values	Moderate	Ignores signs
SD	All values	Sensitive (squared)	Best for algebra
CV	SD and mean	Sensitive	Unit-free; needs a positive mean

! Common mistakes

- × Dividing by n for a sample variance.
- × Reporting variance in the original units.
- × Comparing SDs of data in different units (use CV).
- × Forgetting to multiply by c in the grouped shortcut.

? Fill in the blanks

1. Variety A has SD 4 q and mean 40 q; variety B has SD 3 q and mean 20 q. The more consistent variety has CV = _____ %. [10 (variety A)]
2. If every observation is doubled, the SD is multiplied by _____ and the variance by _____. [2; 4]
3. If yield is recorded in kg, its variance is expressed in _____. [kg²]
4. The observations 5, 5, 5, 5 have a standard deviation of _____. [0]
5. With Q_1 = 18 and Q_3 = 30, the quartile deviation is _____. [6]
6. Adding 10 kg to every observation changes the SD by _____. [nothing (0)]

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Skewness and Kurtosis

Measuring the asymmetry and tail behaviour of a distribution with quantiles and moments.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 Skewness: the idea

Skewness is the degree of asymmetry of a distribution.

Shape	Tail	Typical order*
Symmetric	Balanced	Mean = Median = Mode
Positive	Longer **right** tail	Mean > Median > Mode
Negative	Longer **left** tail	Mean < Median < Mode

*For many unimodal distributions; not a universal rule.

2 Karl Pearson's coefficient

Using the mode

$$S_k = \frac{\text{Mean} - \text{Mode}}{\text{SD}}$$

Using the median

$$S_k = \frac{3(\text{Mean} - \text{Median})}{\text{SD}}$$

Based on the mean, the mode (or median) and the standard deviation. Sign gives the direction: $+$ positive, $-$ negative, 0 symmetric.

3 Bowley's (quartile) measure

Based on quartiles

$$S_Q = \frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1}$$

Equivalent: $\frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 - Q_2) + (Q_2 - Q_1)}$. Lies between −1 and +1; usable with open-end classes.

4 Kelley's (percentile) measure

Based on P_{10}, P_{50}, P_{90}

$$S_P = \frac{(P_{90} - P_{50}) - (P_{50} - P_{10})}{(P_{90} - P_{50}) + (P_{50} - P_{10})}$$

Uses more of the distribution than Bowley's middle 50%.

5 Central moments

rth central moment

$$\mu_r = \frac{\sum f_i (x_i - \bar{x})^r}{N}$$

μ_0	μ_1	μ_2	μ_3	μ_4
1	0 (always)	Variance σ^2	Skewness	Kurtosis

6 Moment skewness

Size only

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

Size and direction

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}}$$

$\gamma_1 = 0$ means zero third-moment skewness; it does not prove symmetry.

7 Kurtosis

Based on the **fourth** standardised central moment; describes tail behaviour and extreme observations.

Kurtosis

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

Excess kurtosis

$$\gamma_2 = \beta_2 - 3$$

Type	β_2	γ_2	Example
Mesokurtic	= 3	= 0	Normal distribution
Leptokurtic	> 3	> 0	Heavier tails, sharper peak
Platykurtic	< 3	< 0	Lighter tails, flatter

! Common mistakes

- × β_1 has no sign; use γ_1 for direction.
- × The skewed mean–median–mode order is typical, not guaranteed.
- × "Positive skew" means the long tail is on the **right**.

? Fill in the blanks

1. Mean 52, mode 46 and SD 12 give Karl Pearson's coefficient S_k = _____. [0.5]
2. Mean 30, median 33 and SD 6 give S_k = _____, so the distribution is _____ skewed. [−1.5; negatively]
3. If Q_1 = 10, Q_2 = 15 and Q_3 = 20, Bowley's coefficient of skewness is _____. [0]
4. A distribution with β_2 = 4.2 is called _____. [leptokurtic]
5. If μ_2 = 4 and μ_4 = 48, then β_2 = _____, so the curve is mesokurtic. [3]
6. β_1 cannot show the direction of skewness because it is never _____. [negative]

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Measures of Association

Scatter diagrams, covariance, Pearson's and Spearman's correlation, Kendall's concordance and correlation matrices.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 Relationships & the scatter diagram

Linear
Can be represented by a straight line.

Monotonic
One variable consistently rises (or falls) as the other rises, not necessarily at a constant rate.

A **scatter diagram** plots paired observations (x_i, y_i): always look at it before computing a coefficient.

2 Types of correlation

Positive	Move in the same direction
Negative	Move in opposite directions
Simple	Two variables
Multiple	One variable with a set of others
Partial	Two variables, controlling for others
Linear / non-linear	Straight-line / curved pattern

3 Karl Pearson's correlation coefficient r

Covariance

$$cov(x,y) = \frac{1}{n}\sum(x_i - \bar{x})(y_i - \bar{y})$$

Correlation

$$r = \frac{cov(x,y)}{sd(x)\, sd(y)}$$

Deviation form

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

Measures **linear** association. Covariance depends on units; r is unit-free. r is sensitive to outliers.

4 Reading r

+1	Perfect positive
near +1	Strong positive
0	No **linear** correlation
near −1	Strong negative
−1	Perfect negative

$-1 \le r \le +1$. Perfect: all points on a line.

5 Spearman's rank correlation ρ

Computed from **ranks**; measures **monotonic** association. Good for ordinal data and curved but monotonic patterns. Range −1 to +1.

No tied ranks

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

d_i = difference between the ranks of the ith pair. Ties get their **average rank** and correction terms $T = \frac{1}{12}\sum m(m^2-1)$.

6 Kendall's coefficient of concordance W

Measures **agreement among several judges' rankings** (e.g. k judges ranking n varieties).

Concordance

$$W = \frac{12\left[\sum R_i^2 - \frac{(\sum R_i)^2}{n}\right]}{k^2 n(n^2-1)}$$

R_i = sum of ranks for object i; k = number of judges.

7 Correlation matrix & correlogram

Correlation matrix
Table of pairwise correlations among several variables. Diagonal elements = **1**; symmetric about the diagonal.

Correlogram
A graphical display of the correlation matrix (colour or size shows strength and sign).

	Pearson r	Spearman ρ
Uses	Actual values	Ranks
Detects	Linear association	Monotonic association
Outliers	Sensitive	Less sensitive

! Common mistakes

- × Correlation is **not causation**.
- × $r = 0$ rules out only **linear** association; a curve can still exist.
- × Skipping the scatter plot.
- × Forgetting average ranks for ties.

? Fill in the blanks

1. If cov(x, y) = 12, sd(x) = 3 and sd(y) = 5, then r = _____. [0.8]
2. If all points lie exactly on a falling straight line, r = _____. [−1]
3. For n = 10 pairs with Σd² = 33, Spearman's ρ = _____. [0.8]
4. Two judges who rank five varieties in exactly the same order give ρ = _____. [1]
5. Changing yield from kg to quintals changes r by _____. [nothing (r is unit-free)]
6. r = 0 rules out a straight-line relationship but not a _____ one. [curved (non-linear)]

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5-MINUTE REVISION

Regression Analysis

Fitting the least-squares line, regression coefficients, the two regression lines and model assumptions.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 The model

Regression models how a **dependent** (response) variable depends on one or more **independent** (predictor) variables.

Simple linear regression

$$y_i = \alpha + \beta x_i + e_i$$

Error e_i

Theoretical deviation from the true line

Residual $\hat{e}_i$

Observed – fitted: $y_i - (\hat{\alpha} + \hat{\beta} x_i)$

2 Slope & intercept

- **Slope** β: expected change in y for a one-unit increase in x. Range $-\infty$ to $+\infty$; $+$ direct, $-$ inverse.
- **Intercept** α: value of y when $x = 0$.

Simple	1 predictor
Multiple	2 or more: $y = \alpha + \beta_1 x_1 + \cdots + \beta_k x_k + e$
Logistic	Categorical response

3 Least squares

Choose α, β to **minimise** $E = \sum [y_i - (\alpha + \beta x_i)]^2$. This gives the **normal equations**:

First

$$\sum y = n\alpha + \beta \sum x$$

Second

$$\sum xy = \alpha \sum x + \beta \sum x^2$$

Slope

$$\hat{\beta}_{yx} = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

Intercept

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

Equivalently $\hat{\beta}_{yx} = cov(x, y)/var(x)$. The fitted line passes through $(\bar{x}, \bar{y})$.

4 Two regression lines

y on x (predict y)

$$\beta_{yx} = r \frac{\sigma_y}{\sigma_x}$$

x on y (predict x)

$$\beta_{xy} = r \frac{\sigma_x}{\sigma_y}$$

Link to correlation

$$\beta_{yx}\, \beta_{xy} = r^2$$

5 Properties of regression coefficients

- ✓ Both have the **same sign**, and r takes that sign: $r = \pm\sqrt{\beta_{yx}\beta_{xy}}$.
- ✓ If one exceeds 1, the other must be less than 1.
- ✓ $r = 0$: both coefficients are zero.
- ✓ $r = \pm 1$: the two lines **coincide**.
- ✓ Independent of change of **origin**, not of **scale**: with $X = \frac{x-a}{h}$, $Y = \frac{y-b}{k}$, $\beta_{YX} = \frac{h}{k}\beta_{yx}$.

6 Assumptions

- **Linearity** of the mean relationship
- **Independence** of errors (no **autocorrelation**)
- Constant error variance: **homoscedasticity**
- **Normality** of errors, for exact inference
- No exact **multicollinearity** among predictors
- Correct model specification

Check the fitted model with residual plots.

7 Correlation vs regression

	Correlation	Regression
Purpose	Measures association	Models the relationship and predicts
Symmetry	Symmetric: $r_{xy} = r_{yx}$	Asymmetric: y on $x \neq x$ on y
Units	Unit-free	Coefficients carry units
Range	−1 to +1	Unbounded

! Common mistakes

- × Using the y-on-x line to predict x.
- × Coefficients of opposite signs (impossible).
- × Treating a regression fit as proof of cause.
- × Predicting far outside the data range.

? Fill in the blanks

1. With ŷ = 5 + 2x, the predicted y at x = 10 is ———. [25]
2. If x̄ = 4, ȳ = 13 and the slope is 2, the intercept is ———. [5]
3. If b_yx = 0.8 and b_xy = 0.45, then r = ———. [0.6]
4. If b_yx = 1.6, then b_xy cannot be more than ———. [0.625]
5. If b_yx is negative, b_xy must also be ———. [negative]
6. When r = 0, the two regression lines are ——— to each other. [perpendicular]

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Probability

Random experiments, events, the three approaches to probability, addition and multiplication laws, and Bayes' theorem.

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1 Basic terms

Random experiment	Outcome cannot be predicted with certainty
Sample space S	Set of all possible outcomes
Sample point	One possible outcome
Event	A subset of the sample space
Random variable	Real-valued function of outcomes: discrete or continuous

2 Types of events

Mutually exclusive	Cannot occur together
Exhaustive	Union is the whole sample space
Equally likely	Equal probability
Independent	$P(A \cap B) = P(A)P(B)$
Complement A^c	All outcomes outside A
Impossible / certain	$P = 0$ / $P = 1$

3 Three approaches

Approach	Definition
Classical (mathematical)	$P(E) = \frac{m}{n}$: favourable ÷ total, for exhaustive, mutually exclusive, equally likely outcomes
Statistical (relative frequency)	$P(E) = \lim_{n\to\infty} \frac{m}{n}$ from repeated trials
Axiomatic (Kolmogorov)	(1) $P(E) \geq 0$ · (2) $P(S) = 1$ · (3) additivity for mutually exclusive events

Always $0 \leq P(E) \leq 1$ and $P(A^c) = 1 - P(A)$.

4 Set notation

$A \cup B$	A or B or both
$A \cap B$	Both A and B
A^c	Not A
$A \mid B$	A given B

5 Addition law

Any two events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Mutually exclusive

$$P(A \cup B) = P(A) + P(B)$$

6 Conditional & multiplication

Conditional, $P(B) > 0$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

General

$$P(A \cap B) = P(A)P(B \mid A)$$

Independent

$$P(A \cap B) = P(A)P(B)$$

With replacement: draws independent. Without replacement: generally dependent.

7 Bayes' theorem

Gives the probability of a **cause** given the observed **evidence**.

Two events

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Partition $E_1, \ldots, E_n$

$$P(E_i \mid A) = \frac{P(E_i)P(A \mid E_i)}{\sum_k P(E_k)P(A \mid E_k)}$$

Prior

Probability before the evidence: $P(E_i)$

Posterior

Updated probability after the evidence: $P(E_i \mid A)$

8 Counting

Combinations (order ignored)

$${}^nC_r = \frac{n!}{r!\,(n-r)!}$$

Gambler's fallacy: in independent trials, past outcomes do not change the next probability.

! Common mistakes

- × Mutually exclusive ≠ independent: exclusive events with non-zero probabilities are always **dependent**.
- × Forgetting to subtract $P(A \cap B)$ for events that can occur together · Confusing $P(A \mid B)$ with $P(B \mid A)$.

? Fill in the blanks

1. If P(A) = 0.3, P(B) = 0.5 and A and B are independent, P(A ∩ B) = _____. [0.15]
2. If P(A) = 0.4, P(B) = 0.5 and P(A ∩ B) = 0.2, then P(A ∪ B) = _____. [0.7]
3. If P(A ∩ B) = 0.12 and P(B) = 0.3, then P(A | B) = _____. [0.4]
4. A seed lot has 90% germination; the probability that a seed fails to germinate is _____. [0.1]
5. The number of ways to choose 2 varieties out of 5 is _____. [10]
6. Two mutually exclusive events, each with positive probability, cannot be _____. [independent]

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Probability Distributions

Expected value, and the Bernoulli, binomial, Poisson and normal distributions with their fitting.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 Distributions & expectation

Discrete

p.m.f. $p(x) = P(X = x)$; $p(x) \geq 0$, $\sum p(x) = 1$

Continuous

p.d.f. $f(x)$; probability = **area** under the curve; total area = 1

Expected value (discrete)

$$E(X) = \sum x\, p(x)$$

Expected value (continuous)

$$E(X) = \int x f(x)\, dx$$

Expected value = the long-run average of the random variable.

2 Bernoulli

A single trial with two outcomes: success (1), failure (0). Parameter p.

p.m.f.

$$p(x) = p^x (1-p)^{1-x}, \quad x = 0, 1$$

Mean	p	Variance	$p(1-p)$

3 Binomial $B(n, p)$

Number of successes in n **fixed, independent** Bernoulli trials with **constant** success probability p; $q = 1 - p$.

p.m.f.

$$p(x) = \binom{n}{x} p^x q^{n-x}$$

Mean · Variance

$$E(X) = np, \quad Var(X) = npq$$

Fitting: estimate p

$$\hat{p} = \frac{\bar{x}}{n}$$

Variance < mean. Expected frequency = $N \times p(x)$, $x = 0, 1, \ldots, n$.

4 Poisson $P(\lambda)$

Counts of events (0, 1, 2, ...) at a rate; approximates rare binomial successes.

p.m.f.

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Mean = Variance = λ. Fit with $\hat{\lambda} = \bar{x}$.

5 Normal $N(\mu, \sigma^2)$

p.d.f.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- ✓ Continuous, bell-shaped, symmetric about μ; mean = median = mode
- ✓ Skewness 0, kurtosis $\beta_2 = 3$; odd central moments are zero
- ✓ Tails approach the x-axis but never touch it
- ✓ Q.D. ≈ 0.6745σ; M.D. ≈ 0.7979σ

6 Standard normal & areas

Standardise

$$z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

Interval	Area
$\mu \pm \sigma$	68.26%
$\mu \pm 1.96\sigma$	95%
$\mu \pm 2\sigma$	95.44%
$\mu \pm 2.58\sigma$	99%
$\mu \pm 3\sigma$	99.73%

7 Compare the distributions

	Binomial	Poisson	Normal
Type	Discrete	Discrete	Continuous
Parameters	n, p	λ	μ, σ
Mean	np	λ	μ
Variance	npq	λ	σ^2
Agri example	Germinated seeds out of 10	Insects per leaf	Plant height

Z-table: area to the left of z. Above z: $1-$ table value. Between two z's: subtract.

! Common mistakes

- × Binomial needs a **fixed** n and **constant** p.
- × $P(X = x) = 0$ for continuous variables: use intervals.
- × Forgetting $1-$ table value for upper-tail areas.

? Fill in the blanks

1. A binomial variable with n = 10 and p = 0.3 has mean _____ and variance _____. [3; 2.1]
2. If 80% of seeds germinate, the expected number germinating out of 20 is _____. [16]
3. A Poisson variable with variance 4 has mean _____. [4]
4. Plant height ~ N(100, 10^2): a plant of 120 cm has z = _____. [2]
5. About _____ % of a normal population lies between μ − 2σ and μ + 2σ. [95.44]
6. In a binomial distribution the variance is always _____ than the mean. [less]

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Sample Survey

Survey terminology, sampling and non-sampling errors, confidence intervals, and probability and non-probability sampling methods.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 Key terms

Census	Complete enumeration of every unit
Sampling unit	Basic unit selected
Sampling frame	Complete list identifying the units
Sampling design	Procedure used to select the sample
Estimator / estimate	Statistic used for a parameter / its particular value
Unbiased	$E(t) = \theta$; e.g. $E(\bar{X}) = \mu$

Sampling fraction

$$f = \frac{n}{N}$$

2 Errors in surveys

Sampling error

Arises from observing a sample, not the whole population. **Absent in a census**. Larger n generally reduces standard error.

Non-sampling error

Everything else: non-response, measurement, recording, processing. Present even in a **census**.

Reliable results need good design, representative sampling, trained enumerators and careful execution.

3 Confidence interval

An **interval estimate** of a parameter.

Confidence level

$$100(1-\alpha)\%$$

$\alpha = 0.05$ → 95%; $\alpha = 0.01$ → 99%. "95%" describes the long-run performance of the method.

4 Probability sampling methods

Method	How	Best when
Simple random (SRSWR / SRSWOR)	Equal inclusion probabilities, with or without replacement	Complete frame, fairly homogeneous population
Stratified	Split into non-overlapping **strata**; sample each independently	Strata internally homogeneous: better precision
Systematic	Random start, then every kth unit; $k = N/n$	Ordered lists, field rows
Cluster	Select clusters at random; **enumerate all** units in them	No unit-level frame; dispersed population
Multistage	Sample in stages (e.g. village → field → plot)	Large-scale surveys such as GCES

Probability sampling allows probability-based inference and measurable sampling error.

5 Non-probability sampling

- **Convenience:** easily accessible units
- **Consecutive:** every accessible eligible unit in turn
- **Quota:** fill subgroup quotas non-randomly
- **Purposive (judgement):** deliberately chosen informative units
- **Snowball:** participants refer others

Useful without a frame or for hard-to-reach groups, but sampling error cannot be measured by probability theory.

6 Survey steps & pioneers

Principal steps: objectives → target population → frame and units → data needed → questionnaire → collection method → **pilot survey** → design and sample size → fieldwork (handle non-response) → edit, tabulate, analyse → report → document.

A.N. Kiaer	Representative Method, 1895
A.L. Bowley	Random sampling and probability-based inference in surveys
P.C. Mahalanobis	Pioneered large-scale sample surveys in India

! Common mistakes

× Calling a convenience sample "random" · A census still has non-sampling errors · SRS draws are independent only **with** replacement · Stratified (sample within every group) ≠ cluster (take whole groups).

? Fill in the blanks

1. Selecting 50 farms out of 1000 gives a sampling fraction of _____. [0.05]
2. To choose 20 plots systematically from 400, the sampling interval k = _____. [20]
3. Splitting a district into irrigated and rainfed areas and sampling from each is _____ sampling. [stratified]
4. Choosing some villages at random and surveying every farmer in them is _____ sampling. [cluster]
5. Wrongly recorded answers can occur even in a census; they are _____ errors. [non-sampling]
6. Interviewing only the farmers who came to a field day is _____ sampling. [convenience]

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Hypothesis Testing

Null and alternative hypotheses, Type I and II errors, significance and power, critical regions, tails and p-values.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 Hypotheses

A **statistical hypothesis** is a testable statement about a population parameter or distribution.

H_0: null

No effect, no difference, no relationship. The one tested for possible rejection.

H_1 / H_a: alternative

An effect, difference or relationship. Decides one- or two-tailed.

Karl Popper's **falsification principle**: a scientific hypothesis must be testable and potentially falsifiable.

2 Errors in decisions

Decision	H_0 true	H_0 false
Reject H_0	Type I error (α)	Correct: power $1-\beta$
Fail to reject	Correct ($1-\alpha$)	Type II error (β)

Level of significance

$$P(\text{reject } H_0 \mid H_0 \text{ true}) \le \alpha$$

Power

$$1-\beta$$

Larger samples generally increase power. α is fixed **before** the test (usually 0.05 or 0.01).

3 Regions & tails

Test statistic (Z, t, χ^2, F) → compared with its **sampling distribution** under H_0. The **critical value** separates the **rejection (critical) region** from the **acceptance region**.

H_1	Test	Reject H_0 when	α in tail
$\mu \neq \mu_0$	**Two-tailed**	Statistic below the lower or above the upper critical value	$\alpha/2$ each
$\mu > \mu_0$	**Right-tailed**	Statistic > critical value	α
$\mu < \mu_0$	**Left-tailed**	Statistic < critical value	α

4 The p-value

Probability, assuming H_0 true, of a statistic **as extreme or more extreme** than observed.

p ≤ α	Reject H_0
p > α	Fail to reject H_0

Small p = strong evidence against H_0.

5 The eight steps of a test

1. State H_0
2. State H_1
3. Set the significance level α
4. Collect the data
5. Calculate the test statistic
6. Find the critical value or region
7. Compare the statistic with it
8. Make and state the decision

6 History

Fisher's tea-tasting experiment: a lady claimed she could tell whether milk or tea was poured first. It's the classic illustration of significance testing.

! Common mistakes

× Saying "accept H_0": say **fail to reject** · Failing to reject does not prove H_0 true · Choosing α after seeing the data · Picking the tail from the result instead of from H_1.

? Fill in the blanks

1. Declaring a new fertiliser effective when it actually has no effect is a Type ____ error. [I]
2. If β = 0.2, the power of the test is ____. [0.8]
3. Testing whether a variety yields more than 40 q/ha needs a ____-tailed test. [right]
4. With p-value 0.03 and α = 0.05, we ____ H_0. [reject]
5. With p-value 0.08 and α = 0.05, we ____ H_0. [fail to reject]
6. In a two-tailed test at α = 0.05, each tail has area ____. [0.025]

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Large Sample Tests

The Central Limit Theorem and Z tests for proportions, means and standard deviations.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 Central Limit Theorem

As n increases

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$$

- Large sample: generally $n \geq 30$.
- Needs **independent** observations and **finite variance**.
- **Standard error** = SD of a sampling distribution; for $\bar{x}$ it is $\sigma/\sqrt{n}$, which shrinks as n grows.

2 Z test decision rules

Test	Reject H_0 if	5%	1%
Two-tailed	$\lvert Z\rvert > Z_{\alpha/2}$	1.960	2.576
Right-tailed	$Z > Z_{\alpha}$	1.645	2.326
Left-tailed	$Z < -Z_{\alpha}$	−1.645	−2.326

State H_0 about the **population parameter**; compute Z from **sample statistics**.

3 Proportions

Single proportion, $H_0 : P = P_0$

$$Z = \frac{p - P_0}{\sqrt{P_0 Q_0 / n}}, \quad Q_0 = 1 - P_0$$

Two proportions, $H_0 : P_1 = P_2$

$$Z = \frac{p_1 - p_2}{\sqrt{\hat{P}\hat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Pooled $\hat{P} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$, $\hat{Q} = 1 - \hat{P}$. Needs enough expected successes **and failures**.

4 Single mean, $H_0 : \mu = \mu_0$

σ known

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

σ unknown

$$Z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

With a large sample, s is a good substitute for σ.

5 Two means, $H_0 : \mu_1 = \mu_2$

Equal SDs

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Pooled σ if unknown

$$\sigma = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}}$$

Unequal SDs

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

For independent samples, the variance of a difference is the **sum** of the variances.

6 Two SDs

Approximate test

$$Z = \frac{s_1 - s_2}{\sqrt{\frac{s_1^2}{2n_1} + \frac{s_2^2}{2n_2}}}$$

Under normal-population assumptions.

7 Choosing the Z test

Question	H_0	Standard error uses
Is the germination rate 80%?	$P = P_0$	P_0 (hypothesised)
Do two districts have the same adoption rate?	$P_1 = P_2$	Pooled $\hat{P}$
Is mean yield 40 q/ha?	$\mu = \mu_0$	σ or s
Do two varieties yield the same?	$\mu_1 = \mu_2$	Common or separate variances

! Common mistakes

- × Using p instead of P_0 in the single-proportion SE.
- × Ignoring the sign of Z in one-tailed tests.
- × Using 1.96 for a one-tailed 5% test (it's 1.645).

? Fill in the blanks

1. With σ = 10 and n = 25, the standard error of the mean is _____. [2]
2. x̄ = 52, μ_0 = 50 and SE = 1 give Z = _____, so a two-tailed test at 5% _____ H_0. [2; rejects]
3. With P_0 = 0.5, n = 100 and p = 0.6, the Z statistic is _____. [2]
4. Making the sample four times larger makes the standard error _____. [half as large]
5. Z = 1.8 in a right-tailed test at 5% means H_0 is _____. [rejected (1.8 > 1.645)]
6. Samples of 100 with p_1 = 0.4 and p_2 = 0.6 give a pooled proportion $\hat{P}$ = _____. [0.5]

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Small Sample Tests

Student's t tests, the correlation test, chi-square tests and the F test for variances.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 Small samples

Often $n < 30$ (not a strict boundary). Uses the exact t, χ^2 and F distributions.

W.S. Gosset published the t distribution as "**Student**". It has heavier tails than the normal.

2 Student's t tests for means

Test	Statistic	df
One sample, $H_0: \mu = \mu_0$	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$	$n - 1$
Two samples, equal variances	$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$, pooled $s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$	$n_1 + n_2 - 2$
Welch (unequal variances)	$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	Welch–Satterthwaite (not an integer)
Paired: before/after on the same unit	$t = \frac{\bar{d}}{s_d/\sqrt{n}}$, $d_i = x_i - y_i$	$n - 1$

3 Testing a correlation

$H_0: \rho = 0$, **df** $= n - 2$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Assumes a bivariate normal model.

4 F test: two variances

$H_0: \sigma_1^2 = \sigma_2^2$

$$F = \frac{s_1^2}{s_2^2}, \quad df = (n_1 - 1,\ n_2 - 1)$$

Put the larger variance on top ($F \geq 1$) and use the upper $\alpha/2$ cut-off for a two-sided test.

5 Chi-square tests

Test	Statistic	df
Goodness of fit	$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$	$n - 1 - m$
Independence ($r \times s$ table)	$\chi^2 = \sum\sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$, $E_{ij} = \frac{A_i B_j}{N}$	$(r-1)(s-1)$
2 × 2 table	$\chi^2 = \frac{n(ad-bc)^2}{C_1 C_2 R_1 R_2}$	1
Population variance	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$	$n - 1$

n categories, m parameters estimated from the data. Expected = row total × column total ÷ grand total. Goodness-of-fit and independence use the **upper** tail.

6 Yates' correction

Optional continuity adjustment for 2 × 2 tables:

Corrected

$$\frac{n[\max(0, |ad - bc| - \frac{n}{2})]^2}{C_1 C_2 R_1 R_2}$$

With small expected counts, consider an exact test.

7 Which test?

Question	Test
Does mean plant height equal 90 cm?	One-sample t
Do two independent varieties differ in yield?	Two-sample t (pooled or Welch)
Did yield change on the same plots before and after a treatment?	Paired t
Do segregation counts fit a 9:3:3:1 ratio?	χ^2 goodness of fit
Is adoption independent of education level?	χ^2 independence
Are two varieties equally variable?	F test

! Common mistakes

- × Two-sample t on paired data.
- × Wrong df: $n - 1$ vs $n_1 + n_2 - 2$ vs $(r-1)(s-1)$.
- × Running χ^2 on percentages instead of counts.

? Fill in the blanks

1. A one-sample t test on 12 plots has _____ degrees of freedom. [11]
2. Two independent groups of 8 and 10 plots (equal variances) give _____ degrees of freedom. [16]
3. Yields of the same 15 plots before and after mulching are compared with the _____ t test. [paired]
4. A 3 × 4 contingency table has _____ degrees of freedom. [6]
5. Row total 40, column total 30 and grand total 100 give an expected frequency of _____. [12]
6. Sample variances 18 and 6 give F = _____. [3]

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Design of Experiments

Treatments, units and factors, experimental error, Fisher's three principles and ways to improve precision.

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1 Why design experiments?

Design of Experiments (DoE): planning experiments so treatment effects are estimated **accurately, efficiently and without bias**.

Objectives: validity · reliability · replicability · optimality.

A good design gives unbiased estimates with their uncertainty, detects important differences, includes the analysis plan, is easy to interpret, has wide validity, uses minimal resources and is as simple as possible.

2 Key terms

Treatment	Any object of comparison
Control	Standard treatment for comparison
Experimental unit	Smallest unit to which a treatment is **independently** applied
Observational unit	Unit on which the response is measured
Response	Measured outcome
Factor / levels	Variable studied / its settings

3 Experimental error

Variation among experimental units **receiving the same treatment**, caused by **extraneous factors**.

Every design must provide a way to **estimate** it.

4 Fisher's three basic principles

Principle	What it means	Why
Randomization	Treatments assigned to units purely by chance	Protects against allocation bias; balances unknown sources of variation on average
Replication	Same treatment on more than one unit	**Estimates experimental error**; improves precision and reliability
Local control (blocking)	Group similar units into blocks before randomising	**Reduces** experimental error from known variation

In a **Randomized Block Design**, all treatments are randomly assigned within each block of similar units.

5 Improving precision

- ✓ **Border effect:** neighbouring plots affect border plants, so record only the inner **net plot**.
- ✓ **Proper plot technique:** keep everything except the treatment uniform.
- ✓ **Appropriate analysis:** e.g. ANCOVA adjusts for covariates.
- ✓ **Planning and recording:** block for time or batches; keep original records.
- ✓ **Matching the design** to the experimental units.

6 History

- **R.A. Fisher** founded the statistical principles of DoE at **Rothamsted Experimental Station** in the 1920s–30s.
- He introduced **ANOVA** to partition variation into factors and chance.
- With **Frank Yates**, he built methods still central to agricultural research.
- Fisher's **tea-tasting** experiment shows why randomization matters.

7 Example: pot experiment with manures

Treatments	Poultry manure, cow dung, coir-pith compost
Experimental unit	One pot receiving one manure
Replication	Several pots per manure
Response	Yield per plant
Error	Yield differences among pots given the same manure

! Common mistakes

- × **Pseudo-replication:** counting several plants in one pot as replicates.
- × Randomizing the field but always applying treatments in the same order.
- × Recording border plants.

? Fill in the blanks

1. Applying each fertiliser dose to four plots uses the principle of _____. **[replication]**
2. Assigning treatments to plots by drawing lots is _____. **[randomization]**
3. Grouping plots of similar fertility into blocks applies _____. **[local control]**
4. Yield differences between plots that received the same treatment measure experimental _____. **[error]**
5. Counting three plants in one pot as three replicates is the mistake called _____. **[pseudo-replication]**
6. Border rows are left out and yield is recorded from the _____ plot. **[net]**

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Uniformity Trials

Studying soil heterogeneity: fertility maps, serial correlation, strip mean squares, Smith's law and optimum plot size.

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1 What is a uniformity trial?

A trial to study the **nature and extent of fertility variation** in a field.

- One crop variety over the whole field; all inputs applied **uniformly**.
- Remove a border, then harvest separately by **basic units** (smallest recorded unit).
- Yield differences among basic units measure **soil heterogeneity**.

Helps decide: plot size, plot shape, block size and blocking direction.

2 Fertility contour map

A map of yields over the field. **Moving averages** smooth random variation; areas of similar fertility are joined by contour lines.

Rice example: 12 m × 17 m field, 1 m border removed → 10 m × 15 m → 150 basic units of 1 m × 1 m.

3 Serial correlation

Neighbouring units

$$r_s = \frac{\sum X_i X_{i+1} - \frac{(\sum X_i)^2}{n}}{\sum X_i^2 - \frac{(\sum X_i)^2}{n}}$$

Computed horizontally and vertically. High r_s: neighbouring yields are similar. Low r_s: little linear association at that lag (not proof of patches).

4 Mean square between strips

Vertical strips

$$MS_V = \frac{\frac{\sum V_i^2}{r} - \frac{G^2}{n}}{c-1}$$

Horizontal strips

$$MS_H = \frac{\frac{\sum H_i^2}{c} - \frac{G^2}{n}}{r-1}$$

A simpler method with the same aim as serial correlation. The larger mean square shows the direction of greater variation. $n = r \times c$.

5 Fairfield Smith's variance law

Variance law

$$V_x = \frac{V_1}{x^b}$$

Log form (linear)

$$\log V_x = \log V_1 - b \log x$$

Regression through origin

$$Y = cX, \quad c = -b$$

V_x = variance of yield per basic unit for plots of x units; b = **Smith's index of soil heterogeneity**, idealised range 0 to 1. $Y = \log V_x - \log V_1$, $X = \log x$.
Example: $b = 0.3607$, adjusted ≈ 0.4116.

6 Simulated plots

Variance per unit area

$$V_x = \frac{Q_x}{rc - x}$$

Combine basic units into plots of x units ($w = rc/x$ plots, fitting the field exactly). A plot's yield is the **sum** of its units.

7 Maximum curvature method

Plot **plot size** (X-axis) against **coefficient of variation** (Y-axis). CV generally falls as plot size rises; the point of **maximum curvature** (the bend) is taken as the optimum plot size.

Larger plots
Lower CV, but fewer plots fit a fixed area.

Caution
The bend depends on axis scales and the available arrangements: treat it as a guide.

! Common mistakes

- × Different shapes partition the same field, so their variances are **dependent**: no ordinary F test.
- × High b alone doesn't prove a patch pattern.

? Fill in the blanks

1. If Smith's index b = 0, a larger plot _____ the variance per unit area. **[does not reduce]**
2. If b = 1, doubling the plot size _____ the variance per unit area. **[halves]**
3. The slope of log V_x plotted against log x equals _____. **[–b]**
4. A 10 m × 15 m net area cut into 1 m² units gives _____ basic units. **[150]**
5. In the maximum curvature method, CV is plotted against _____. **[plot size]**
6. In a uniformity trial every basic unit gets the _____ variety and management. **[same]**

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5-MINUTE REVISION

Analysis of Variance

Partitioning total variation, degrees of freedom, mean squares, the F test, assumptions and ANOVA models.

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1 The idea

ANOVA (Analysis of Variance) compares population means by **partitioning the total variation** into parts due to known sources and to error.

Null hypothesis (one-way)

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_v$$

A significant F means **at least one pair** of means differs. It is an **omnibus** test and doesn't say which pairs.

2 Assumptions

- ✓ Observations from **normal** populations
- ✓ Common variance: **homogeneity of variance**
- ✓ Samples drawn **independently**
- ✓ Treatment effects are **additive**

So errors are iid $N(0, \sigma^2)$. Check with residual plots: residuals vs fitted values, and a normal Q-Q plot.

3 One-way ANOVA (CRD)

Correction factor

$$CF = \frac{G^2}{n}$$

Total SS

$$SS_T = \sum y_{ij}^2 - CF$$

Treatment SS

$$SS_{Tr} = \sum \frac{T_i^2}{n_i} - CF$$

Error SS

$$SS_E = SS_T - SS_{Tr}$$

Mean squares

$$MS = \frac{SS}{df}$$

F ratio

$$F = \frac{MS_{Tr}}{MSE}$$

v treatments, n observations, T_i treatment totals, G grand total.

4 Degrees of freedom

Source	One-way	Two-way
Treatments	$v-1$	$v-1$
Blocks	–	$r-1$
Error	$n-v$	$(v-1)(r-1)$
Total	$n-1$	$vr-1$

5 Two-way ANOVA (RBD)

Block SS (B_j block totals)

$$SS_B = \sum \frac{B_j^2}{v} - CF$$

Error SS

$$SS_E = SS_T - SS_{Tr} - SS_B$$

Separates treatment and block variation. With one observation per cell, interaction cannot be separated from error.

6 ANOVA models

Linear model

$$Y_{ij} = \mu + \tau_i + e_{ij}, \quad e_{ij} \sim N(0, \sigma^2)$$

Fixed	τ_i unknown constants (specific varieties of interest)
Random	$\tau_i \sim N(0, \sigma_\tau^2)$; variance components (randomly chosen fields)
Mixed	Both fixed and random effects

7 The ANOVA table

Source	df	SS	MS	F
Treatments	$v-1$	SS_{Tr}	$SS_{Tr}/(v-1)$	MS_{Tr}/MSE
Error	$n-v$	SS_E	MSE	
Total	$n-1$	SS_T		

Reject H_0 if calculated F > table F at ($v-1$, error df). Then use multiple comparisons to find which means differ.

! Common mistakes

- × Concluding **all** means differ from a significant F.
- × Forgetting to subtract the correction factor.
- × Deleting outliers just to change significance.

? Fill in the blanks

1. A CRD with 5 treatments and 4 replications has ——— error df. [15]
2. An RBD with 5 treatments in 4 blocks has ——— error df. [12]
3. MS(treatment) = 40 and MSE = 8 give F = ———. [5]
4. A grand total of 120 from 12 observations gives a correction factor of ———. [1200]
5. A treatment SS of 60 with 4 treatments gives a treatment mean square of ———. [20]
6. A significant F shows that at least ——— pair of means differs. [one]

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Multiple Comparison Tests

Why post-hoc tests are needed, and Fisher's LSD (CD), Duncan's DMRT and Tukey's HSD with letter grouping.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 Why a post-hoc test?

A significant ANOVA (F) shows that **not all means are equal**, but not **which pairs** differ.

Number of pairs

$$N = \binom{v}{2} = \frac{v(v-1)}{2}$$

Inflated Type I error

$$1 - (1-\alpha)^N$$

FWER (family-wise error rate): probability of at least one Type I error across the whole family of comparisons.

2 Fisher's LSD = Critical Difference (CD)

Equal replications

$$\text{LSD} = t_{\alpha/2,\,edf}\sqrt{\frac{2\,MSE}{r}}$$

Unequal: $SE(d)$

$$\sqrt{MSE\left(\frac{1}{r_i} + \frac{1}{r_j}\right)}$$

Two steps: significant ANOVA first (**protected LSD**), then pairs. Means differ if $|\bar{T}_i - \bar{T}_j| > \text{LSD}$. Relatively **liberal**.

3 Duncan's Multiple Range Test (DMRT)

SE of a mean

$$SE(\bar{Y}) = \sqrt{\frac{MSE}{r}}$$

Shortest significant range

$$R_p = r_p \cdot SE(\bar{Y})$$

1. Rank the means
2. Get $SE(\bar{Y})$ and table r_p
3. Compare each difference with its R_p

p = number of ordered means spanned, including both ends (adjacent: $p = 2$). R_p grows with p. Popular in plant sciences.

4 Tukey's HSD

Honestly significant difference

$$\text{HSD} = q_{\alpha,(v,\,edf)}\sqrt{\frac{MSE}{r}}$$

Uses the **studentised range** q. Compares all pairs while **controlling the FWER**. One critical value for equal replication. Most **conservative** of the three; needs no preliminary ANOVA.

5 Compare the tests

	LSD / CD	DMRT	Tukey HSD
Critical value	One (t)	Varies with p (r_p)	One (q)
Error control	Good only for 2–3 treatments	Between the two	Controls FWER
Nature	Most liberal	More powerful than Tukey	Most conservative
Worked example	1.10	1.10 – 1.21	1.59

In the example, LSD and DMRT found every pair significant; Tukey did not declare differences of 1.5.

6 Other procedures

- **Dunnett:** every treatment vs one control
- **Bonferroni:** adjusts α for the number of tests
- **Scheffé:** all linear contrasts
- **SNK:** Student–Newman–Keuls

7 Presenting results: letter grouping

Treatment	Mean	Group
T3	24.6	**a**
T1	23.9	**ab**
T2	22.8	**b**

Same letter = not significantly different; **no common letter** = significantly different. A mean can carry two letters (overlapping groups). Line/underscore notation conveys the same information. (Illustrative values.)

! Common mistakes

- × Running many t tests without adjustment.
- × Unprotected LSD after a non-significant F.
- × Misreading "ab" as different from "a".

? Fill in the blanks

1. With 6 treatments there are _____ pairwise comparisons. [15]
2. At α = 0.05, the chance of at least one false positive in 3 independent comparisons is about _____. [0.14]
3. MSE = 8, r = 4 and t = 2.12 give an LSD of _____. [4.24]
4. Means marked “a” and “ab” are _____ different. [not significantly]
5. Comparing every new variety with one check variety uses _____ test. [Dunnett's]
6. Of LSD, DMRT and Tukey's HSD, the most conservative is _____. [Tukey's HSD]

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Single Factor Experiments

Completely Randomized Design, Randomized Complete Block Design and Latin Square Design: layout, ANOVA and CD.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 Single factor

Only **one factor** (the treatments) is varied; all other conditions are kept uniform.

CRD ↔ one-way ANOVA · RCBD ↔ two-way · Latin square ↔ three-way (rows, columns, treatments).

2 The three designs at a glance

	CRD	RCBD	Latin square (LSD)
Principles	Randomization, replication	+ local control (1 direction)	+ local control (2 directions)
Best for	Homogeneous units: lab, greenhouse	One known gradient, e.g. fertility	Two perpendicular gradients
Layout	Fully random	Blocks = replications; every treatment once per block	Each treatment once per row and column; v^2 plots
Error df	$v(r-1) = n - v$	$(v-1)(r-1)$	$(v-1)(v-2)$

For the same v and r, CRD has the **largest** error df, because nothing is set aside for blocking.

3 CRD

Model

$$y_{ij} = \mu + \tau_i + e_{ij}$$

Source	df
Treatments	$v-1$
Error	$n-v$
Total	$n-1$, with $n = rv$

Allows unequal replication. Rarely used in the field, where plots are seldom homogeneous.

4 RCBD

Model

$$y_{ij} = \mu + \tau_i + \beta_j + e_{ij}$$

Source	df
Blocks	$r-1$
Treatments	$v-1$
Error	$(v-1)(r-1)$

Blocks run **across** the gradient so that each block is uniform. Randomize independently within each block.

5 Latin Square Design

Model

$$y_{ijk} = \mu + \rho_i + \gamma_j + \tau_k + e_{ijk}$$

Source	df
Rows / Columns / Treatments	$v-1$ each
Error	$(v-1)(v-2)$
Total	v^2-1

CF $= G^2/v^2$. Suited to a modest number of treatments: too few leave little error df, too many need many homogeneous plots.

6 Critical difference (CD)

CRD (equal) & RCBD

$$t_{\alpha/2,\,edf}\sqrt{\frac{2\,MSE}{r}}$$

Latin square

$$t_{\alpha/2,\,edf}\sqrt{\frac{2\,MSE}{v}}$$

CRD (unequal r_i, r_j)

$$t_{\alpha/2,\,edf}\sqrt{MSE\left(\frac{1}{r_i}+\frac{1}{r_j}\right)}$$

Used only after a significant treatment F. Means sharing a letter are not significantly different.

7 How many replications?

A target of about **12 error df** is a practical planning guide, not a rule. Replication should also reflect the power and precision wanted, the expected variability and the available resources.

Correction factor = (grand total)² ÷ total number of observations. Treatment comparisons always use the **error mean square**.

! Common mistakes

- × Blocks laid **along** the gradient instead of across it.
- × Using RCBD error df for a CRD.
- × Applying CD when F is not significant.

? Fill in the blanks

1. A Latin square for 5 treatments needs ——— plots. [25]
2. A 5 × 5 Latin square has ——— error df. [12]
3. In an RCBD with 6 treatments and 4 blocks, each block has ——— plots. [6]
4. A pot trial in a uniform greenhouse is best laid out as a ———. [CRD]
5. If fertility falls from north to south, blocks should run ———. [east–west (across the gradient)]
6. For the same treatments and replications, ——— has the largest error df. [CRD]

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5-MINUTE REVISION

Factorial Experiments

Main effects, simple effects and interactions; 2ⁿ and 3ⁿ series, Yates' method and confounding.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 The idea

A **factorial experiment** studies two or more factors **simultaneously**, using all combinations of their levels.

Treatment combinations

$$N = \prod s_i$$

Series

$$2^n,\ 3^n$$

$2^2 = 4$, $2^3 = 8$, $2^4 = 16$, $3^2 = 9$, $3^3 = 27$ combinations. Treatment df = $N - 1$.

2 Effects

Main effect	Average change from one factor's level change, over the other factors
Simple effect	Effect of one factor at a **particular level** of another
Interaction	Effect of one factor **depends on** the level of another

Parallel mean profiles = no interaction; non-parallel = possible interaction (needs a test).

3 2^2 factorial (r replications)

Main effect of A

$$\frac{(A)_{i=1} - (A)_{i=0}}{2r}$$

SS due to A

$$SS_A = \frac{[(A)_{i=1} - (A)_{i=0}]^2}{4r}$$

A × B interaction

$$\frac{(AB)_{i+j=0} - (AB)_{i+j=1}}{2r}$$

A, B and A×B each have **1 df**. In 2^3: divisors become $4r$ (effects) and $8r$ (SS); A, B, C, AB, AC, BC, ABC each have 1 df. **Yates' method** is the tabular algorithm for 2^n effects; Fisher's algebraic method is the alternative.

4 Degrees of freedom

Factor with a levels	$a - 1$
A × B	$(a-1)(b-1)$
3^2: A, B	2 each
3^2: A × B	4

A 3-level factor's 2 df split into **linear** $A_L = A_2 - A_0$ and **quadratic** $A_Q = A_2 + A_0 - 2A_1$.

5 Types of factorials

Symmetrical	All factors have the same number of levels (2^n, 3^n)
Asymmetrical	Unequal levels, e.g. 2×3
Complete (full)	All combinations included
Fractional	Only a selected fraction of combinations

6 Critical differences

Interaction means (2^2; ABC in 2^3)

$$t\sqrt{\frac{2\,MSE}{r}}$$

Two-factor means in 2^3

$$t\sqrt{\frac{MSE}{r}}$$

Levels of A in 3^2

$$t\sqrt{\frac{2\,MSE}{3r}}$$

Levels of B in 2×3

$$t\sqrt{\frac{2\,MSE}{2r}}$$

$t = t_{\alpha/2,\,edf}$. If the interaction is significant, compare combination means, not main-effect means.

7 Confounding

Deliberately making an interaction **inseparable from block effects** so that blocks can be smaller.

Complete
The **same** interaction is confounded in every replication.

Partial
Different interactions confounded in different replications.

GAIN smaller, more homogeneous blocks and better precision · **COST** lost information on the confounded effect. Higher-order interactions are usually the ones confounded.

! Common mistakes

- × Interpreting main effects when the interaction is significant.
- × Mixing up $2r$ (effect) and $4r$ (SS) divisors in 2^2.
- × Confounding a main effect.

? Fill in the blanks

1. Three nitrogen levels × two phosphorus levels give ____ treatment combinations. [6]
2. A 2^4 factorial has ____ treatment combinations. [16]
3. In a 3 × 4 factorial, the A × B interaction has ____ df. [6]
4. If the response to nitrogen changes with the irrigation level, the two factors show ____. [interaction]
5. In a 2^3 factorial, each main effect and interaction has ____ df. [1]
6. Confounding ABC in every replication is ____ confounding. [complete]

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Analysis of Covariance

Combining ANOVA with regression: covariates, the common-slope model, adjusted means and the adjusted F test.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 What is ANCOVA?

ANCOVA combines **analysis of variance** with **regression adjustment** on a **covariate**: a measured explanatory variable.

Agri examples: initial plant stand, initial weight of animals, pre-treatment pest count, previous-season yield.

A covariate must be **related to the response** and **not caused by the treatment**. Ideally measured **before** treatment (pre-treatment covariate).

2 Blocking vs ANCOVA

	Blocking	ANCOVA
When	Before the experiment (layout)	During analysis
Handles	Predictable variation (soil, gradient)	Variation in a measured covariate
Aim	Both reduce experimental error	

Precision may improve but is **not guaranteed**.

3 Model & key quantities (CRD)

Common-slope model

$$Y_{ij} = \mu + \tau_i + \beta(X_{ij} - \bar{X}) + e_{ij}$$

Slope

$$\hat{\beta} = \frac{E_{xy}}{E_{xx}}$$

Adjusted error SS

$$E = E_{yy} - \frac{E_{xy}^2}{E_{xx}}$$

Adjusted MSE

$$MSE_{adj} = \frac{E}{n - v - 1}$$

E_{xx}, E_{xy}, E_{yy} = within-treatment (error) sums of squares and products. RBD error df: $(r-1)(v-1) - 1$.

4 Degrees of freedom

Fitting one covariate slope **uses one error df**.

CRD	$n - v - 1$
RBD	$(r-1)(v-1) - 1$

5 Adjusted means & F test

Adjusted treatment mean

$$\bar{Y}_{i,adj} = \bar{Y}_i - \hat{\beta}(\bar{X}_i - \bar{X})$$

Adjusted F

$$F = \frac{(E_1 - E)/(v-1)}{MSE_{adj}}$$

E_1: residual SS of the reduced model (no treatments, covariate kept); E: full model. Means are compared at a **common covariate value**.

6 Assumptions

- ✓ The usual ANOVA assumptions: independent errors, common variance, normality for exact tests
- ✓ **Linear** relationship between response and covariate
- ✓ **Equal slopes** across treatments (check treatment × covariate interaction)
- ✓ Covariate not affected by the treatments
- ✓ Error mean is zero

7 Steps of the analysis

1. Compute X, Y and XY SS for
2. Get error E_{xx}, E_{xy}, E_{yy}
3. Estimate $\hat{\beta} = E_{xy}/E_{xx}$
4. Adjusted error SS and MSE_{adj}
5. Adjusted treatment SS: $E_1 - E$
6. Adjusted F test
7. Compute adjusted means
8. Compare with pair-specific SEs

! Common mistakes

- × Using a covariate the treatment affects.
- × Forgetting the extra df lost.
- × Assuming adjustment turns observational data into an experiment.

? Fill in the blanks

1. Initial plant stand counted before applying treatments is a suitable _____. [covariate]
2. With E_xy = 30 and E_xx = 60, the slope b = _____. [0.5]
3. Treatment mean 50, its covariate mean 12, overall covariate mean 10 and b = 0.5 give an adjusted mean of _____. [49]
4. A CRD with 20 observations, 4 treatments and one covariate has _____ error df. [15]
5. A covariate must not be _____ by the treatments. [affected]
6. The common-slope model assumes the treatment regression lines are _____. [parallel]

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Missing Plot Analysis

Handling a lost observation in CRD, RCBD and Latin square: direct fitting, Bartlett's covariance method and substitution.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 Why plots go missing

Animal or pest damage, flooding, theft, recording errors... Always **record why** an observation is missing.

A poor result **caused by a treatment** is real data and must not be discarded.

A missing plot **destroys the balance** of an RCBD or Latin square; a CRD copes directly.

2 CRD: fit directly

Analyse the observed responses with unequal treatment sample sizes (no substitution needed).

Error df

$$edf = n_{obs} - v$$

SE of a difference

$$SE = \sqrt{MSE\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$

3 RCBD: one missing value

Substitution estimate

$$\hat{x} = \frac{rB + vT - G}{(r-1)(v-1)}$$

B, T = observed totals of the affected block and treatment; G = observed grand total (all **excluding** the missing value).

Error df

$$(r-1)(v-1) - 1$$

4 Latin square: one missing value

Substitution estimate

$$\hat{x} = \frac{v(R + C + T) - 2G}{(v-1)(v-2)}$$

R, C, T = observed totals of the affected row, column and treatment; G = observed grand total.

Error df

$$(v-1)(v-2) - 1$$

5 Bartlett's covariance method

Create an imaginary **covariate** X: 1 (or −1) for the missing plot, 0 elsewhere. Set $Y = 0$ for the missing plot. Then run **ANCOVA** for the design.

The indicator accounts for the missing position: total and error df are each **one less** than in the complete design.

6 After substituting

- ✓ The estimate **minimises the error SS**.
- ✓ Subtract **one** error df per estimated value.
- ✓ The treatment SS can be **inflated**: apply the bias correction.
- ✓ Comparisons involving the affected treatment need adjusted SEs.
- ✓ Several missing values: estimate by **iteration**.

7 Choosing an approach

Design	Approach	Error df
CRD	Fit observed data (unequal n_i)	$n_{obs} - v$
RCBD	Substitution $\hat{x}$, Bartlett's method, or fit the observed data	$(r-1)(v-1)-1$
Latin square	Substitution $\hat{x}$ or Bartlett's method	$(v-1)(v-2)-1$

! Common mistakes

- × Counting $\hat{x}$ as a real observation.
- × Including the missing cell in B, T or G.
- × Ignoring missingness linked to the response itself (bias).

? Fill in the blanks

1. In an RBD with r = 4 and v = 5, B = 60, T = 50 and G = 400 give a missing value estimate x̂ = _____. [7.5]
2. An RBD with 4 treatments in 5 blocks and one missing plot has _____ error df. [11]
3. A 4 × 4 Latin square with one missing plot has _____ error df. [5]
4. The substituted value is the one that makes the error sum of squares _____. [as small as possible]
5. In a CRD, a lost plot is handled by analysing with _____ replications. [unequal]
6. In Bartlett's method the dummy covariate is _____ for the missing plot and 0 elsewhere. [1]

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Data Transformation

When ANOVA assumptions fail: log, square root, angular and Box-Cox transformations, and back-transformation.

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

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1 Why transform?

Used when ANOVA assumptions fail, typically **unequal error variances (heteroscedasticity)**, non-normal errors or non-additive (multiplicative) effects.

1. Transform the data
2. Run ANOVA and mean separation on the **transformed** scale
3. **Back-transform** means to the original units for reporting

Never pick a transformation just to get the conclusion you want.

2 Which transformation?

Data	Variance	Use
Counts (Poisson)	$\sigma^2 = \mu$ or $c^2\mu$	**Square root**
Binomial percentages	$\sigma^2 = p(1-p)/n$	**Angular** (arcsine)
SD ∝ mean; multiplicative effects	$\sigma^2 = c^2\mu^2$	**Logarithmic**

3 Logarithmic

Positive data

$$X = \log Y$$

With zeros

$$X = \log(Y + 1)$$

Back-transform: $Y = \text{antilog}\, X$. Logs need positive inputs. Example in the book: weed counts.

4 Square root

Counts

$$X = \sqrt{Y}$$

Zeros present

$$X = \sqrt{Y + 1}$$

Very small counts: $\sqrt{Y} + \sqrt{Y+1}$. Back-transform $\sqrt{Y+1}$ with $Y = X^2 - 1$. Insect, larval, defective and diseased-seed counts.

5 Angular (arcsine) transformation

Transform

$$X = \sin^{-1}\sqrt{p}, \quad p = \frac{Y}{100}$$

0% and 100%

$$p = \frac{1}{4n}, \quad p = 1 - \frac{1}{4n}$$

Back-transform

$$Y = 100 \sin^2 X$$

For percentages from **binomial counts** (germination, mortality). The 30–70% guideline is approximate. Nutrient percentages (protein, carbohydrate) are not binomial and need not be transformed.

6 Box-Cox

A **family of power transformations** indexed by λ, for strictly positive data.

$\lambda = \frac{1}{2}$: square root · $\lambda = -1$: reciprocal · $\lambda \to 0$: log. The angular transformation is **not** in this family.

7 After transforming

- ✓ Check assumptions using the **residuals** of the fitted model.
- ✓ Use one transformation consistently for the whole dataset.
- ✓ CD on the transformed scale: $t_{\alpha/2,\,edf}\sqrt{2\,MSE/r}$ (RBD).
- ✓ Back-transformed means are **not** arithmetic means (e.g. squared root means are ≤ the original mean).

! Common mistakes

- × Reporting transformed means as final results.
- × Taking log of zero.
- × Arcsine on non-binomial percentages.

? Fill in the blanks

1. Counts of insects per plant are usually analysed after a _____ transformation. [square root]
2. Percent germination out of 100 seeds calls for the _____ transformation. [angular (arcsine)]
3. When the SD rises in proportion to the mean, use the _____ transformation. [logarithmic]
4. A transformed mean √(Y + 1) = 3 back-transforms to _____. [8]
5. An angular value of 30° back-transforms to _____ %. [25]
6. Counts that include zeros are transformed as √(Y + _____). [1]

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